SCENTINEL

OBP panel selection · specificity & discrimination report

iGEM URV 2026 · Generated: 17/10/2026 23:19

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Target VOC      :  methyl salicylate  (CAS 119-36-8)
Starting OBP    :  DcitOBP3
Interferents (5):  r-limonene, L-Linalool, hexanal, benzaldehyde,
                   1-hexanol
Vineyard profile:  mesa_interferentes_real_48hpi.csv
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Tool 1 noise floor   :  0.05
Predicted min. conf. :  0.3
Tool 2 max. error    :  5%  (J >= 5.41)
Tool 2 sensor noise  :  5%
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Verdict of every run

Tool 1 · experimental	Tool 1 · predicted	Tool 2 · experimental	Tool 2 · predicted
WEAK	**CONDITIONAL**	**SOLID**	**SOLID**
3 OBPs	3 OBPs	3 OBPs	3 OBPs

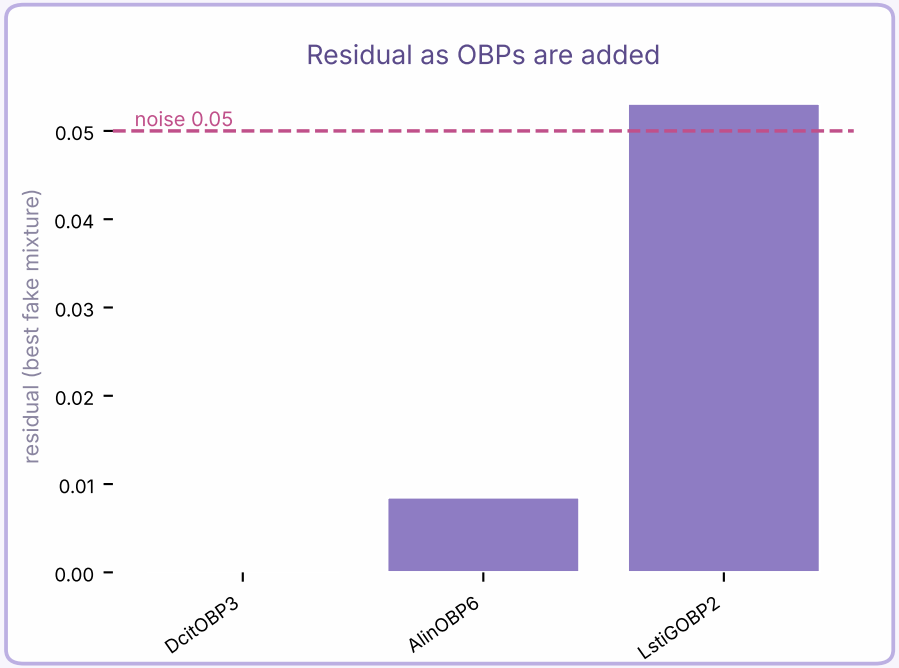
SOLID = no single unverified value can break it · CONDITIONAL = rests on unverified values, predictions agree

WEAK = a bet goes against the prediction (measure first) · FAILS / NOT ENOUGH = does not reach the limit

Tool 1 · Specificity · Experimental data

Smallest OBP panel such that no mixture of the interferents imitates methyl salicylate · 34 candidate OBPs of 103 with target data (24 blind removed) · priority: measured

#	OBP	added by	residual	Ki tgt (uM)	c2	meas/imp	covers
1	DcitOBP3	start	0.0000	7.01	0.40	3/0	
2	AlinOBP6	search	0.0085	33.00	0.80	5/0	
3	LstiGOBP2	search	0.0531	55.00	0.80	5/0	



row tint: c2 >= 0.8 green · 0.5-0.8 lilac · < 0.5 pink · c2 = how much is known about the interferents on that OBP (measured 1, predicted x confidence, missing 0)

VERDICT: WEAK

Panel: DcitOBP3 + AlinOBP6 + LstiGOBP2

Residual 0.0531 vs noise 0.05 → clears the noise floor on the data used.

Why: 2 bet(s) go AGAINST the Tanimoto prediction.

If every not-measured pair took its Tanimoto prediction at once: residual 0.0080 → FAILS.

Closest fake mixture: 0.739 x r-limonene, 0.207 × 1-hexanol

! DcitOBP3 has c2 = 0.40 (< 0.50): little is known about the interferents on it, it may look more specific than it is.

Tool 1 · What the panel bets on · Experimental data

Every value the 'pass' depends on that nobody measured (or measured once and is fragile) · most fragile first

Bets (7)

#	for the panel to work, this must be true	data	risk
1	hexanal binds DcitOBP3 weakly or not at all (Ki > 7.94 uM)	not measured; prediction Ki 3.77 uM, conf 0.47, left out (counted as 'does not bind')	AGAINST prediction
2	L-Linalool binds DcitOBP3 weakly or not at all (Ki > 25.1 uM)	not measured; prediction Ki 3.09 uM, conf 0.59, left out (counted as 'does not bind')	AGAINST prediction
3	Ki of methyl salicylate on DcitOBP3 = 7.01 uM is accurate (fails if it is 4% higher)	measured, single value	lost if off by x1.04
4	Ki of r-limonene on DcitOBP3 = 8.25 uM is accurate (fails if it is 6% lower)	measured, single value	lost if off by x1.06
5	Ki of r-limonene on AlinOBP6 = 11.3 uM is accurate (fails if it is 6% higher)	measured, single value	lost if off by x1.06
6	Ki of 1-hexanol on DcitOBP3 = 9.05 uM is accurate (fails if it is 19% lower)	measured, single value	lost if off by x1.19
7	benzaldehyde binds DcitOBP3 weakly or not at all (Ki > 10 uM)	no measurement, no prediction (counted as 'does not bind')	lost if off by x3

Pairs to measure (5), most urgent first

OBP	compound	status	risk	breaks the panel if	Ki pred (uM)	conf
DcitOBP3	hexanal	not used	BREAKS (prediction)	Ki <= 7.94	3.77	0.47
DcitOBP3	L-Linalool	not used	BREAKS (prediction)	Ki <= 25.1	3.09	0.59
DcitOBP3	benzaldehyde	no data	risk	Ki <= 10	-	-
AlinOBP6	L-Linalool	not used	safe	never	18.1	0.59
LstiGOBP2	r-limonene	not used	safe	never	55	0.50

Tool 1 · Specificity · Predicted data (conf >= 0.3)

Smallest OBP panel such that no mixture of the interferents imitates methyl salicylate · 68 candidate OBPs of 136 with target data (9 blind removed) · priority: score

#	OBP	added by	residual	Ki tgt (uM)	c2	meas/imp	covers
1	DcitOBP3	start	0.0000	7.01	0.61	3/2	
2	PameOBP24	search	0.0101	19.97	0.68	3/3	
3	LmigOBP4	search	0.1546	4.37	0.66	2/3	



row tint: c2 >= 0.8 green · 0.5-0.8 lilac · < 0.5 pink · c2 = how much is known about the interferents on that OBP (measured 1, predicted x confidence, missing 0)

VERDICT: CONDITIONAL

Panel: DcitOBP3 + PameOBP24 + LmigOBP4

Residual 0.1546 vs noise 0.05 → clears the noise floor on the data used.

Why: it rests on unverified values, but each has a 2x margin or more.

If every not-measured pair took its Tanimoto prediction at once: residual 0.1546 → still passes.

Closest fake mixture: 1.364 × 1-hexanol

Tool 1 · What the panel bets on · Predicted data (conf >= 0.3)

Every value the 'pass' depends on that nobody measured (or measured once and is fragile) · most fragile first

Bets (4)

#	for the panel to work, this must be true	data	risk
1	methyl salicylate binds LmigOBP4 at least this strongly: Ki < 12.6 uM	not measured; prediction Ki 4.37 uM, conf 0.48, used	lost if off by x2.88
2	1-hexanol binds LmigOBP4 weakly or not at all (Ki > 3.55 uM)	not measured; prediction Ki 13.6 uM, conf 0.79, used	lost if off by x3.82
3	r-limonene binds LmigOBP4 weakly or not at all (Ki > 7.08 uM)	no measurement, no prediction (counted as 'does not bind')	lost if off by x4.24
4	L-Linalool binds LmigOBP4 weakly or not at all (Ki > 2.51 uM)	not measured; prediction Ki 14.6 uM, conf 0.50, used	lost if off by x5.8

Pairs to measure (10), most urgent first

OBP	compound	status	risk	breaks the panel if	Ki pred (uM)	conf
LmigOBP4	methyl salicylate	imputed	risk	Ki >= 12.6	4.37	0.48
LmigOBP4	1-hexanol	imputed	risk	Ki <= 3.55	13.6	0.79
LmigOBP4	r-limonene	no data	risk	Ki <= 7.08	-	-
LmigOBP4	L-Linalool	imputed	risk	Ki <= 2.51	14.6	0.50
DcitOBP3	benzaldehyde	no data	safe	never	-	-
DcitOBP3	hexanal	imputed	safe	never	3.77	0.47
DcitOBP3	L-Linalool	imputed	safe	never	3.09	0.59
PameOBP24	1-hexanol	imputed	safe	never	10	0.42
PameOBP24	L-Linalool	imputed	safe	never	6.81	0.50
PameOBP24	hexanal	imputed	safe	never	4.29	0.50

Tool 1 · Experimental vs predicted data

Mode 1 picks the best-MEASURED useful OBPs; mode 2 lets the Tanimoto predictions take part in the choice

Experimental data

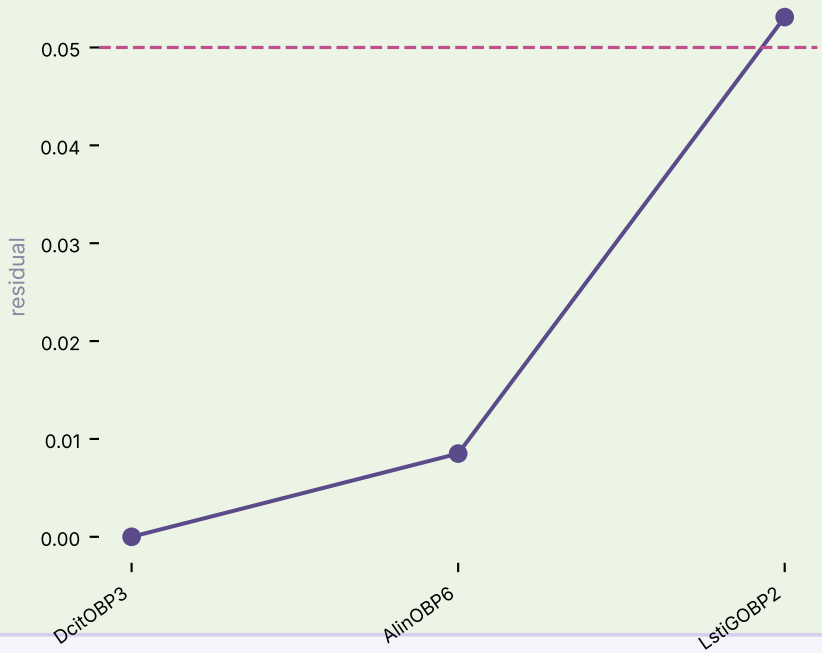
WEAK

Panel (3): DcitOBP3 + AlinOBP6 + LstiGOBP2

Residual: 0.0531 (noise 0.05)

Bets: 7 against the prediction: 2

Pairs without a lab Ki: 5



Predicted (Tanimoto) data

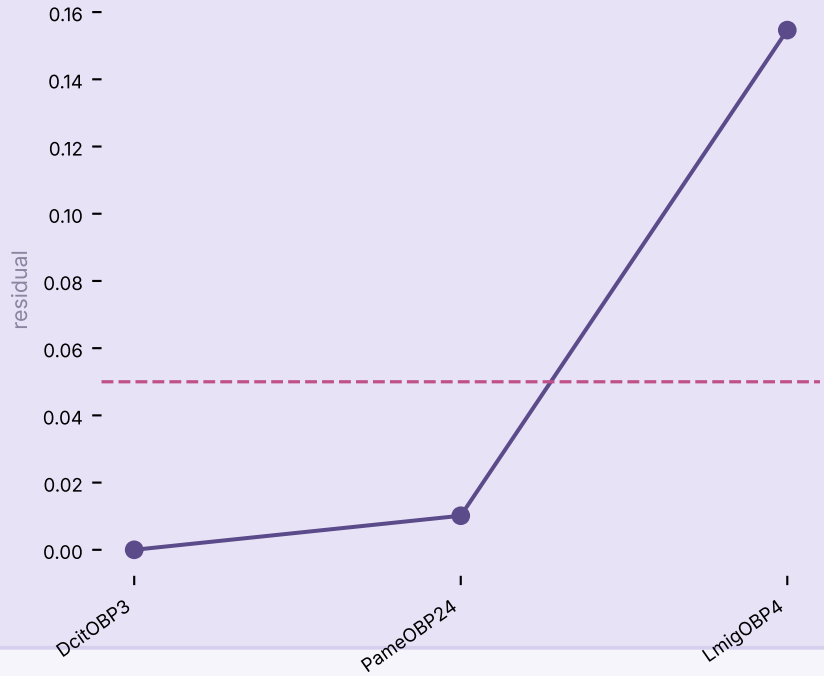
CONDITIONAL

Panel (3): DcitOBP3 + PameOBP24 + LmigOBP4

Residual: 0.1546 (noise 0.05)

Bets: 4 against the prediction: 0

Pairs without a lab Ki: 10

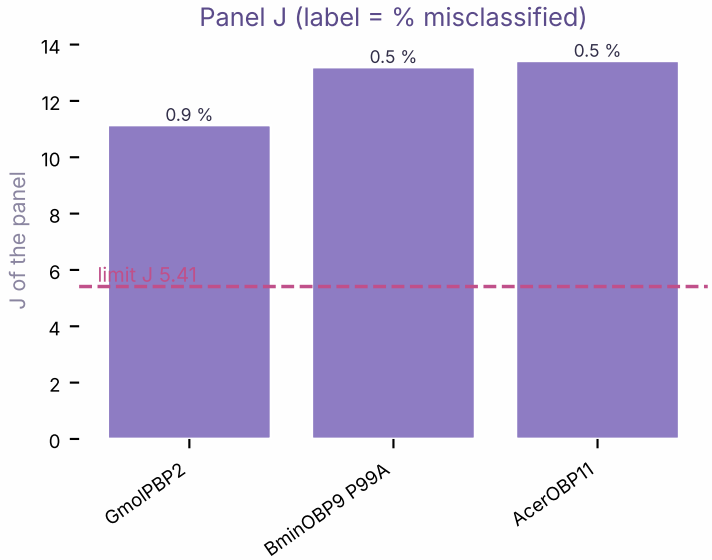


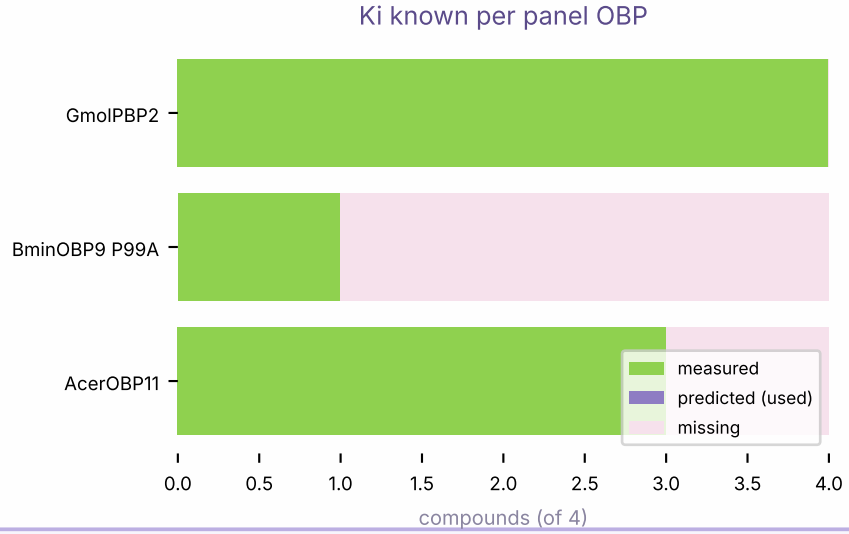
Tool 2 · Discrimination (healthy vs diseased) · Experimental data

4 compounds, mean ± SD (n = 3 per group) · 133 candidate OBPs · J = Fisher ratio, err = plants misclassified = Φ(−√(J/2))

Best single OBPs

OBP	J	err	Ki meas.	reading healthy	reading diseased	binds (Ki uM)
GmolPBP2	11.14	0.9 %	4/4	203.5 ± 78.7	929.3 ± 200.7	methyl salicylate(7.5), 1-hexanol(10.4), benzaldehyde(10.5)
GmolOBP16	11.04	0.9 %	4/4	84.8 ± 31.7	386.1 ± 84.2	1-hexanol(17.5), methyl salicylate(20.2), benzaldehyde(23.8)
GmolOBP7	11.01	0.9 %	4/4	159.8 ± 64.8	754.3 ± 165.4	methyl salicylate(9.8), benzaldehyde(14.1), DL linalool(15.2)
AlinCSP4	10.85	1.0 %	3/4	210.3 ± 79.2	920.8 ± 198.6	1-hexanol(7.9), methyl salicylate(8.2), benzaldehyde(9.1)
GmolOBP8	10.84	1.0 %	3/4	93.2 ± 36.6	421.6 ± 91.9	methyl salicylate(18.8), benzaldehyde(20.9), 1-hexanol(21.6)
AlinCSP5	10.72	1.0 %	3/4	127.5 ± 58.8	690.7 ± 160.3	methyl salicylate(9.7), 1-hexanol(17.3), benzaldehyde(21.5)
DcitOBP6	10.52	1.1 %	3/4	64.6 ± 35.5	417.0 ± 102.0	DL linalool(6.1), methyl salicylate(16.0), 1-hexanol(21.2)
PxylGOBP2	10.45	1.1 %	3/4	125.5 ± 62.8	769.6 ± 187.8	DL linalool(4.5), 1-hexanol(5.8), methyl salicylate(8.0)





VERDICT: SOLID

Panel: GmolPBP2 + BminOBP9 P99A + AcerOBP11

J = 13.42 (0.5 % misclassified); limit J >= 5.41 (max 5% of plants misclassified).

Why: no single unmeasured Ki can plausibly pull J under the limit.

All Tanimoto predictions at once: J = 11.45 (0.8 %) → within the limit.

Compound shift (diseased − healthy, SDs): 1-hexanol +1.1, benzaldehyde +1.1, methyl salicylate +3.0, DL linalool +1.5

Tool 2 · What the panel bets on · Experimental data

Each unmeasured Ki moved over 0.1 uM - 10 mM (one at a time, and each compound on all panel OBPs at once)

Bets that can pull J under the limit: 0

No unmeasured Ki can pull J under the limit on its own, nor any compound on all panel OBPs at once.

The table below shows how far J could fall in the worst case.

Largest possible drops of J (all scanned Ki, worst first)

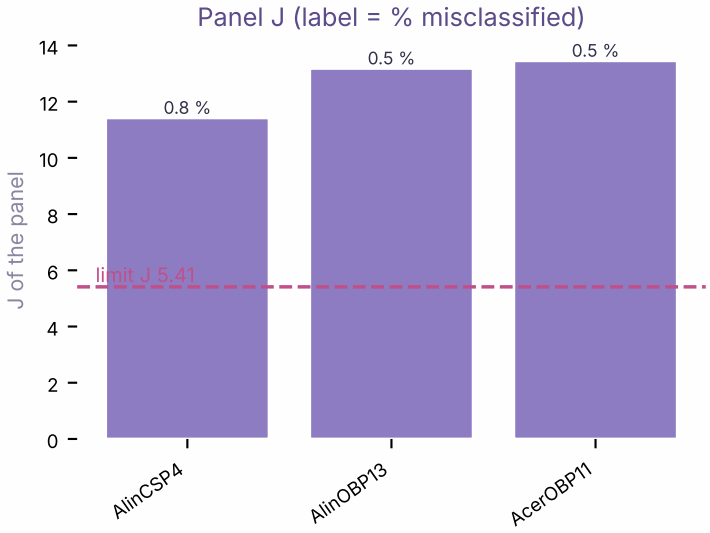
OBP	compound	status	Ki pred (uM)	worst J	misclassified	verdict
ALL 2: BminOBP9 P99A+AcerOBP11	benzaldehyde	joint	-	11.21	0.9 %	safe
BminOBP9 P99A	methyl salicylate	no data	-	11.38	0.9 %	safe
BminOBP9 P99A	1-hexanol	not used	1.56	11.49	0.8 %	safe
BminOBP9 P99A	benzaldehyde	no data	-	11.57	0.8 %	safe
AcerOBP11	benzaldehyde	not used	4.16	13.20	0.5 %	safe

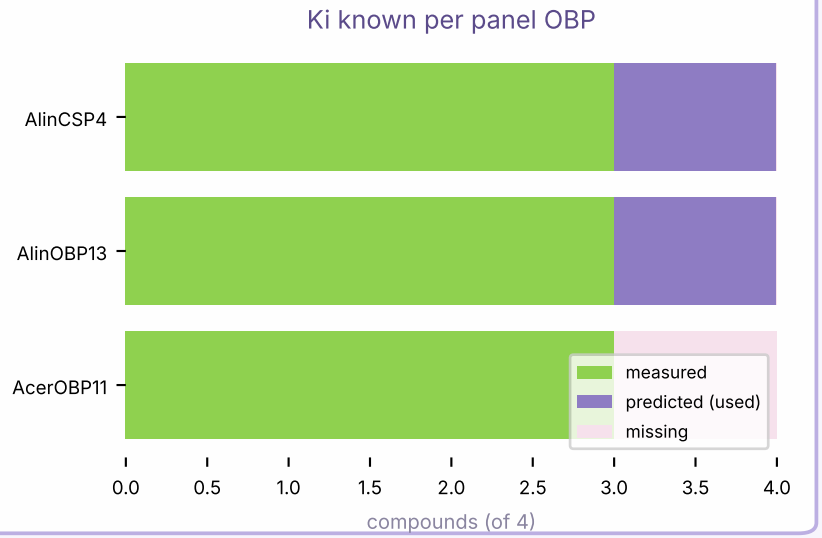
Tool 2 · Discrimination (healthy vs diseased) · Predicted data (conf >= 0.3)

4 compounds, mean ± SD (n = 3 per group) · 282 candidate OBPs · J = Fisher ratio, err = plants misclassified = Φ(−√(J/2))

Best single OBPs

OBP	J	err	Ki meas.	reading healthy	reading diseased	binds (Ki uM)
AlinCSP4	11.42	0.8 %	3/4	222.0 ± 79.2	952.8 ± 199.1	DL linalool(2.3), 1-hexanol(7.9), methyl salicylate(8.2)
GmolPBP2	11.14	0.9 %	4/4	203.5 ± 78.7	929.3 ± 200.7	methyl salicylate(7.5), 1-hexanol(10.4), benzaldehyde(10.5)
GmolOBP16	11.04	0.9 %	4/4	84.8 ± 31.7	386.1 ± 84.2	1-hexanol(17.5), methyl salicylate(20.2), benzaldehyde(23.8)
GmolOBP7	11.01	0.9 %	4/4	159.8 ± 64.8	754.3 ± 165.4	methyl salicylate(9.8), benzaldehyde(14.1), DL linalool(15.2)
BmorCSP2	10.99	1.0 %	0/4	515.0 ± 189.3	2198.6 ± 466.3	1-hexanol(0.9), benzaldehyde(0.7), methyl salicylate(0.7)
SexiOBP1	10.97	1.0 %	0/4	191.4 ± 71.7	822.0 ± 174.5	benzaldehyde(9.9), methyl salicylate(9.9), DL linalool(10.5)
AlinCSP5	10.92	1.0 %	3/4	130.6 ± 58.8	699.1 ± 160.3	methyl salicylate(9.7), 1-hexanol(17.3), DL linalool(18.8)
GmolOBP8	10.84	1.0 %	3/4	93.2 ± 36.6	421.6 ± 91.9	methyl salicylate(18.8), benzaldehyde(20.9), 1-hexanol(21.6)





VERDICT: SOLID

Panel: AlinCSP4 + AlinOBP13 + AcerOBP11

J = 13.45 (0.5 % misclassified); limit J >= 5.41 (max 5% of plants misclassified).

Why: no single unmeasured Ki can plausibly pull J under the limit.

All Tanimoto predictions at once: J = 13.19 (0.5 %) → within the limit.

Compound shift (diseased − healthy, SDs): 1-hexanol +1.1, benzaldehyde +1.1, methyl salicylate +3.0, DL linalool +1.5

Tool 2 · What the panel bets on · Predicted data (conf >= 0.3)

Each unmeasured Ki moved over 0.1 uM - 10 mM (one at a time, and each compound on all panel OBPs at once)

Bets that can pull J under the limit: 0

No unmeasured Ki can pull J under the limit on its own, nor any compound on all panel OBPs at once.

The table below shows how far J could fall in the worst case.

Largest possible drops of J (all scanned Ki, worst first)

OBP	compound	status	Ki pred (uM)	worst J	misclassified	verdict
ALL 2: AlinCSP4+AlinOBP13	DL linalool	joint	-	11.25	0.9 %	safe
AlinOBP13	DL linalool	imputed	16.6	11.66	0.8 %	safe
AcerOBP11	benzaldehyde	not used	4.16	13.18	0.5 %	safe
AlinCSP4	DL linalool	imputed	2.25	13.45	0.5 %	safe

Lab priorities

The measurements that decide whether the panels can be trusted (all runs, most urgent first; same pair listed once)

#	OBP	compound	action	why	affects
1	DcitOBP3	hexanal	measure Ki	**AGAINST prediction**	Tool 1 exp.
2	DcitOBP3	L-Linalool	measure Ki	**AGAINST prediction**	Tool 1 exp.
3	DcitOBP3	methyl salicylate	repeat the measurement (replicates)	repeat (fragile value)	Tool 1 exp.
4	DcitOBP3	r-limonene	repeat the measurement (replicates)	repeat (fragile value)	Tool 1 exp.
5	AlinOBP6	r-limonene	repeat the measurement (replicates)	repeat (fragile value)	Tool 1 exp.

Method · how every number is computed and how missing Ki are handled

TOOL 1 - SPECIFICITY

1. Ki -> signal: s = 1/Ki (1/uM). Missing Ki = 0 ('does not bind'). '>X' = 1/(1.1 X) (an interferent is assumed to bind a little: conservative). Predicted mode: target from *_imputed_pred, interferents from *_imputed_lower.
2. Residual of a panel P: min over concentrations c >= 0 of || A c - b || (NNLS). b = target signals on P, A = interferent signals on P. It is the distance between the target and the best possible fake mixture. The panel works when residual > sensor noise (default 0.05 ~ 5 points of occupancy at target 1 uM).
3. Candidates: OBPs with target data, c2 >= 0.5 (c2 = interferent knowledge: measured 1, predicted x confidence, missing 0, / number of interferents), and not blind (only '>X' values for every compound).
4. Search: starts from the OBP you give. Mode 1 (experimental) adds, among the OBPs that raise the residual by >= 0.001, the one with the highest gain x c2^2 (best-measured useful OBP); mode 2 adds the one with the highest residual. Coverage: every interferent must bind (measured, not '>X', or predicted) a panel OBP.
5. Bets: each unmeasured pair is set to every Ki from 0.1 uM to 10 mM; the Ki range that brings the residual under the noise is reported and compared with the Tanimoto prediction. Measured Ki that break the panel with a change <= 25 % are listed as fragile values.

TOOL 2 - DISCRIMINATION

1. Ki -> signal ('anchored', as main.py): s = log10(40/Ki) / log10(40), clipped to 0-1 (Ki <= 1 uM -> 1, Ki >= 40 uM -> 0). Missing Ki and '>X' = 0.
2. Input: mean and SE per compound and class; SD = SE x sqrt(n). Reading of an OBP = sum(s x amount). Mean reading mu = A @ mean; covariance Sig = A diag(SD^2) A^T (compounds vary independently).
3. J = d^T (Sig_healthy + Sig_diseased + N)^-1 d, d = mu_diseased - mu_healthy, N = (5 % of the reading)^2. One OBP: J = d^2 / (var_h + var_d + noise). Misclassified = Phi(-sqrt(J/2)): 5 % -> J >= 5.4.
4. Bets: every unmeasured Ki of the panel (and each compound on all panel OBPs at once) is moved over 0.1 uM - 10 mM. A hidden compound that varies between plants more than it separates the classes lowers J: a missing Ki can make J look too HIGH. A multi-OBP panel can subtract shared noise, so it resists hidden Ki better.

VERDICTS (both tools): SOLID = no single unverified value can plausibly break it; CONDITIONAL = it rests on unverified values but the predictions do not contradict them; WEAK = a bet goes against the prediction, or the panel fails if every prediction is true at once. Full explanation: docs/MISSING_DATA_ES.md.